

Mechanical Problem Statements

Three problems, one theme: the robot only moves the way you designed it to if the mechanism is correctly synthesized, the motion is verified against numbers, and the design is proven, not assumed.



SCAN TO JOIN THE
WHATSAPP GROUP

- 01** **Mechanism Design: Single-DOF Walking Linkage** MEDIUM PAGE 4

Mechanism synthesis & kinematics — discover a single-actuator linkage that walks, verify its foot-tip path against a target trajectory, and prove it never gains a second degree of freedom.
- 02** **Octopus Arm: n -Joint Gripper Mechanism** MEDIUM PAGE 7

Underactuated synthesis & CAD — a many-jointed gripping arm driven by exactly two actuated degrees of freedom, with the coupling relation derived and a physical model that proves it works.
- 03** **Variable-Damping Suspension for a Sample-Search Rover** HARD PAGE 10

Cross-domain system design — a semi-active suspension whose damping responds in real time to terrain sensed ahead of the rover, protecting a sensitive payload on unknown, rocky ground.

How to Approach These Problems

These are not exam questions with a marking key. Every one of them can be answered shallowly, and every one of them rewards going further than the question strictly asked for. We can tell the difference in about two minutes of conversation.

WHAT WE ARE ACTUALLY TESTING

Whether you can be trusted with the mechanism layer of a competition robot. That means dimensions with justification behind them, decisions you can defend, and an honest account of what you derived versus what you guessed. A confident CAD render built on an unverified degree-of-freedom count is worse than no answer.

Several of these ask you to derive something, and several ask you to generate and compare something. Do both properly. A derivation with the working shown and the assumptions written down is worth far more than a final number on its own, because we can see where your reasoning goes when the assumptions change.

IF YOU DO NOT HAVE CAD OR SIMULATION ACCESS

Some of these are better with SolidWorks, Fusion, MATLAB/Simulink or Gazebo in front of you. If you cannot get that, say so clearly and do the work anyway: derive the kinematics by hand, sketch the mechanism at several phases of motion, and be specific about what you predict a simulation would show. A well-reasoned hand derivation you could not simulate beats a vague description of one you did.

USING LLMs

LLMs and AI tools are allowed on every problem here. Use them, to learn a topic quickly or to get unstuck. One condition: you own everything you submit. If a dimension is in your report, you know where it came from and what it means.

Mechanism design punishes shortcuts as much as electrical or software work does. A model will happily hand you a link length, a degree-of-freedom count or a damping coefficient that looks entirely reasonable and is wrong for your case, and nothing will tell you until the assembly does not close in CAD, or the rover bottoms out on the first rock. Check every number against a hand calculation or a simulation you ran yourself.

HOW YOU SUBMIT

One GitHub repository link. No slides. Reports, calculations, diagrams, CAD files, trajectory plots, simulation files, everything goes in there. Hand-drawn diagrams photographed and committed are completely acceptable. Where a problem asks for a demo or a physical result, add a short edited video showing it working.

Single-DOF Walking Linkage Mechanism

Design a leg mechanism that walks on uneven terrain using exactly one degree of freedom and one actuator. The aim is not to look up a known walking linkage and resize it. It is to be the person who can derive, from scratch, why a single rotating input can produce a foot-tip path with a flat stance phase and a clean lift — and only reach for the literature once your own attempts have taught you what to look for.

TYPE	COVERS	PLUS	TARGET DATA
Synthesis, derivation & CAD	Four-bar & multi-bar coupler curves, DOF	Trajectory matching, CAD realization	Team-Deimos-IIT-Mandi/Robotic-PS

1 Why This Comes First

Most arguments between mechanism designers start at the same place: someone adds a joint to fix one problem and quietly changes the degree of freedom of the whole assembly without noticing. The linkage still moves when you push it, so it looks fine on the bench — right up until the actuator is bolted on and the mechanism binds, or moves in a way nobody commanded.

A rotating crank does not, by itself, guarantee a useful foot path. What the coupler point actually traces depends entirely on the lengths and pivot placement you choose, and there is no shortcut around working through that geometry yourself — by hand, by sketch, or in simulation — before you trust a CAD render of it.

2 What to Study

This is the list, not a syllabus to tick off. Some of it you will read in an afternoon, some will take real digging. Go deeper wherever you find you cannot explain something out loud.

DEGREES OF FREEDOM

- Grübler's / Kutzbach equation for planar mechanisms, and how to count links, joints, and joint types correctly — including the mistakes that silently change the count (a joint that looks like one pin but is mechanically two, a link that is redundant, a higher pair mistaken for a lower pair).
- Why a mechanism with one visible input can still have more than one DOF if a sub-loop isn't fully constrained, and how to spot it before you build it.

COUPLER CURVE THEORY

- Four-bar linkages: the Grashof condition, and what it tells you about whether a link can fully rotate versus only oscillate.
- Coupler curves: how a point rigidly attached to the coupler link traces a closed path as the crank completes one revolution, and how changing link-length ratios reshapes that path.
- Six-bar and eight-bar extensions built from stacked four-bar loops sharing a common crank — how adding a second loop off the same input lets you combine two coupler points into a rigid sub-assembly with a new, more useful path.
- Cognate linkages: different link-length combinations that trace the identical coupler curve. Useful once you have a shape you like and want a mechanically cleaner version of it.

WHEN IT GOES WRONG

- Toggle and dead-centre positions: where a linkage momentarily loses effective mechanical advantage or reverses direction unpredictably.
- Branch defects: the same set of link lengths can assemble in more than one configuration, and only one of them is the motion you designed for.
- Interference: a mechanically valid coupler curve that nonetheless requires two links to occupy the same space at some point in the cycle.

3 The Target Path

THE OBJECTIVE

Get your foot-tip path close to the target shape and height without hard-coding a known mechanism's dimensions from a search result.

This is the part of the problem we care most about, because it teaches something reading alone cannot: that a walking linkage's usefulness lives entirely in the shape of one curve, and that curve is sensitive to every link length you chose.

A reference foot-tip trajectory is provided as a set of (x, y) coordinates for one full actuator cycle, together with a plot. It has a long, nearly flat stance-phase segment and a tall, rounded swing-phase loop, peaking at a lift height of 200 mm.

THE TARGET DATA

target_foot_tip_path.csv and target_foot_tip_path.png are in the problem repository: github.com/Team-Deimos-IIT-Mandi/Robotic-PS

Work from the CSV for coordinates. The figure below is indicative shape only.

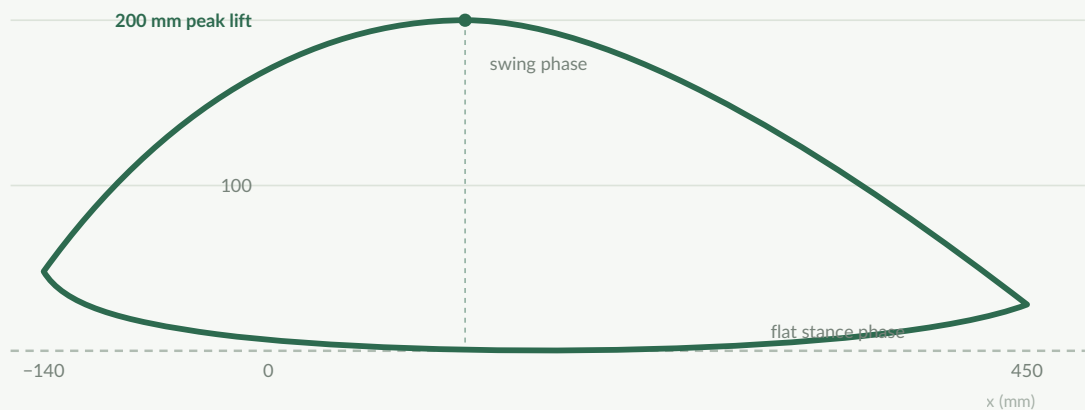


Figure 1. Target foot-tip path for one full actuator cycle: long flat stance segment, swing loop peaking at 200 mm. Indicative shape only — work from the supplied CSV for coordinates.

STEP WHAT YOU'RE TESTING

- 1 First attempt, from first principles.** Pick a linkage topology (four-bar coupler, six-bar, whatever you can justify) and plot its coupler curve without looking anything up. Compare shape by eye against the target.
- 2 Iterate on link lengths.** Adjust dimensions and pivot placement to push your curve toward the target's proportions — the long flat segment and the height of the loop above it.
- 3 Re-derive the DOF.** Every time you add a link or joint to reshape the curve, recheck Grübler's equation. It is very easy to accidentally add a second DOF while chasing a better curve.
- 4 Scale to the target height.** Once the shape is right, scale your whole linkage uniformly so the peak lift matches 200 mm exactly, and report the resulting stride length that comes along with it.
- 5 Only now, research.** If you want to see what the literature calls the family of mechanisms you've arrived at, this is the point to look. Say what you found and how close your independent attempt already was.

You are not expected to derive a mathematically exact curve fit — that is a harder numerical-optimization problem than the timeline here allows. A design that visibly captures the same character (flat stance, high swing loop, correct approximate lift height) is a strong submission on its own.

Each of those is a place error hides if left unstated.

IF YOU HAVE NO CAD OR SIMULATION TOOL

Say so up front. Then do the geometry by hand: sketch the linkage at several crank angles to scale, construct the coupler point position at each, and connect the points into an approximate path. A carefully constructed hand-drawn coupler curve at eight or more crank positions is a completely acceptable substitute.

Name every assumption

Which link is grounded, where the input pivot sits, and why you picked the branch you did.

4 The DOF Proof

Reading Grübler's equation is not the same as applying it to your own assembly without making a mistake. Do this one properly and show the working.

SHOW THE ARITHMETIC

List every link, every joint, and every joint type in your final mechanism, then apply the equation and show the result equals exactly one. An answer with no working shown is not an answer.

5 Adapting the Design

The real test is being asked “what would you change for this terrain” and having an answer with reasons behind it. Work through all three. For each, say which dimension or pivot you'd retune, and what trade-off that creates elsewhere in the cycle.

#	TERRAIN
1	Frequent small rocks and debris, roughly 10–15 mm tall, closely spaced.
2	Smooth, mostly flat gravel — clearance matters less than stride efficiency and speed.
3	A sustained 15° incline, where stance-phase traction and mechanical advantage matter more than lift height.

There is no single right answer to any of the three. There are answers you can defend and answers you cannot.

6 What to Hand In

ITEM	WHAT IT CONTAINS
DOF proof	Section 4, full Grübler's equation working for your final mechanism.
Kinematic diagram	Labeled schematic (hand-drawn or CAD sketch): all links, pin joints, ground link, driven input, key dimensions annotated.
Trajectory plot	Your coupler curve for one full input rotation, overlaid against the target path, with a short note on how close you got and what you'd tune further given more time.
CAD model	3D assembly in at least two distinct phases of motion, mechanically realizable — no interference, reasonable link thickness and joint clearance.
Adapting the design	Section 5, all three terrain scenarios, with reasoning.
Sources	What you read, and at which point in your process you read it — before or after your own first attempt.

QUESTIONS YOU SHOULD EXPECT

Walk us through your Grübler's count, term by term. Why did you pick this branch of the mechanism and not the other one it could have assembled into? Your coupler curve is close in height but the stance phase isn't flat — which link length is responsible? What happens to your lift height if you scale only the crank and leave every other link the same? Where does your mechanism actually interfere with itself, if anywhere, across the full cycle?

Octopus Arm: n -Joint Gripper Mechanism

Design a gripping arm with n joints (or continuum segments), where $n \geq 4$, driven by exactly **two** actuated degrees of freedom. The aim is not to bolt a servo to every joint and call it a gripper. It is to work out, from the coupling geometry itself, how two independent inputs can be distributed across many joints so the arm curls, wraps and grips the way an octopus arm does with no bone and no individually addressable joint.

TYPE	COVERS	PLUS	OUTPUT
Underactuated synthesis, coupling derivation & CAD	Tendon coupling, differentials, underactuation ratios	Grasp-mode comparison, physical sketch model	Coupling proof, joint-angle plots, CAD, model photos

1 Why This Comes Second

Problem statement 01 was about making sure a mechanism has *exactly* the degrees of freedom you intend, no more and no less. This one asks a harder, related question: what happens when you deliberately want *fewer* actuators than joints?

An octopus arm has, for practical purposes, an unbounded number of bending points and only a handful of independent muscle groups driving all of them. Nobody wired a motor to every vertebra. The arm curls in a useful, predictable way because the coupling between muscle groups and joint angles was shaped by the geometry and material distribution of the arm itself — not by having one actuator per joint.

The temptation with a gripper is to reach for n servos and n joints and call it solved. That solves grabbing. It does not solve the actual problem here, which is producing a coherent, compliant wrapping motion across many joints from a low-dimensional input. A rigid multi-link arm with two independent actuation paths, correctly coupled, will out-grip a naively overactuated one on irregular objects — and it will do it with a fraction of the electronics, mass, and failure points.

2 What to Study

This is the list, not a syllabus to tick off. Go deeper wherever you cannot explain something out loud.

UNDERACTUATION

- What it means for a mechanism to have more joints than actuators, and why that is not automatically a degrees-of-freedom violation once you introduce passive elements (springs, elastic tendons, mechanical stops) that constrain the remaining motion.
- The distinction between an underactuated *joint* (passively compliant, springs back) and an underactuated *arm* (globally coupled, every joint's motion depends on every other joint's state through the same actuation path).

COUPLING MECHANISMS

- Tendon-driven coupling: a single cable routed through pulleys or eyelets at each joint, where pulling the cable closes every joint it passes through, at a ratio set by pulley radius and routing offset from each joint's pivot.
- Differential coupling: how a single differential (or a stacked pair) lets two actuators produce independent *bend* and *twist*, or independent bend of two arm halves, rather than one actuator per motion.
- Linkage-based coupling: a slider or four-bar sub-mechanism that mechanically links adjacent joint angles by a fixed ratio, so one input drives a whole chain in a prescribed pattern (constant curvature, or a curvature that changes predictably along the arm).
- Series elastic and antagonistic tendon pairs: two tendons pulling against each other, or against a spring, to give you both actuation *and* some compliance for free.

CONTINUUM AND SOFT-RIGID HYBRID ARMS

- Constant-curvature approximation: treating a multi-segment arm as if it bends into a smooth circular arc, and why this is a useful simplification even for a rigid-link approximation of a continuum arm.
- Backbone approaches: a flexible or segmented spine with a small number of tendons routed along its length, versus a fully rigid n -bar chain with the same tendon routing.

- Where real octopus- and elephant-trunk-inspired robots put their actuators, base-mounted versus distributed, and why base-mounting is almost always preferred for mass and inertia reasons.

GRASP MECHANICS

- Enveloping (power) grasp versus pinch grasp, and why an underactuated coupled arm naturally favours enveloping grasps on irregular or fragile objects — it conforms instead of concentrating force at fingertip points.
- Contact sequencing: which joint should close first as the arm wraps an object, and how your coupling ratio determines that order. A poorly chosen ratio can close the tip before the base ever touches the object.

SUGGESTED VIEWING

Watch these before you commit to a coupling strategy. Seeing how existing tendon- and pneumatic-driven arms actually move is worth more than a paragraph of description. None of them hand you a coupling relation; you still have to derive that yourself.

- **Festo OctopusGripper in action** — pneumatically bent silicone tentacle with two rows of suckers. [youtube.com/watch?v=WeEJRjNpvP0](https://www.youtube.com/watch?v=WeEJRjNpvP0)
- **The Octopus Robot** — a second look at the same gripper family with more close-up shots of the wrap. [youtube.com/watch?v=Q-cdzRmEx9o](https://www.youtube.com/watch?v=Q-cdzRmEx9o)
- **PSI — FESTO Robotic Tentacle** — the gripper mounted on an arm, grasping varied objects. [youtube.com/watch?v=b7vegkm-hoU](https://www.youtube.com/watch?v=b7vegkm-hoU)
- **ROS multi-section continuum-robot visualizer** — how segment count and curvature parameters change a coupler-driven arm's shape before you touch hardware. [youtube.com/watch?v=sLpNf3Wudt8](https://www.youtube.com/watch?v=sLpNf3Wudt8)
- **“Octopus Inspired Robot Arm Can Think for Itself Underwater”** — distributed sensing in a soft arm, and why contact sequencing matters even without joint-level sensing. [youtube.com/watch?v=v0lk7SKGtCI](https://www.youtube.com/watch?v=v0lk7SKGtCI)

3 The Objective

Design a gripping arm of at least $n = 4$ rigid links (more if you want a smoother wrap), connected by revolute joints, in which the entire chain is driven by exactly two independent actuator inputs. The arm should be able to:

1. Curl from a straight, extended configuration into a closed wrap, using one actuation path.
2. Use the second actuation path to do something the first cannot — twist about the arm's own axis, differentially bend two halves of the arm, or independently set the curvature profile along the length. State clearly what the second input controls and why you chose it.

STEP WHAT YOU'RE TESTING

- 1 Pick a coupling strategy.** Tendon-and-pulley, differential, or linkage-coupled. Justify the choice against what the arm needs to grip.
- 2 Derive the coupling relation.** Write the equations mapping actuator displacement to each joint angle. This is the heart of the problem — it replaces the DOF count from problem 01.
- 3 Simulate or hand-plot the curl.** Show joint angles, or the arm's overall shape, at several actuator positions from fully extended to fully wrapped.
- 4 Check contact order.** For a target object diameter of your choosing, show which joint contacts first, second, and so on, and confirm the order is sensible for an enveloping grasp.
- 5 Only now, research.** Compare your coupling to known continuum-arm or underactuated-gripper designs. Say what you found and how close your independent attempt already was.

You are not expected to match a commercial gripper's grasp force or cycle time. A design that visibly and correctly demonstrates two-input control over an n -joint chain, with a defensible coupling relation, is a strong submission on its own.

Each of those is a place error hides if left unstated.

Name every assumption

Tendon anchor point, where the differential sits, how you handled friction and backlash.

IF YOU HAVE NO CAD OR SIMULATION TOOL

Say so up front. Then do the geometry by hand: sketch the arm at several actuator positions to scale, using your derived coupling relation to compute each joint angle by hand, and draw the resulting arm shape at each stage. A carefully constructed hand-drawn sequence at five or more actuator positions is a completely acceptable substitute.

4 The Coupling Proof

This replaces the Grübler's-equation proof from problem 01. Reading about underactuated coupling is not the same as deriving it correctly for your own mechanism. Do this one properly and show the working.

SHOW THE ARITHMETIC

List every joint, every actuator, and every passive element (spring, elastic tendon, mechanical stop) in your final mechanism. Derive the explicit relation $\theta_i = f_i(x_1, x_2)$ for each joint angle as a function of your two actuator inputs, and show that the relation is well-defined — no joint left with an indeterminate angle — across the full range of motion.

5 Adapting the Design

The real test is being asked “what would you change for this object” and having an answer with reasons behind it. Work through all three. For each, say which parameter — tendon routing offset, pulley radius ratio, coupling stiffness, number of active joints — you'd retune, and what trade-off that creates elsewhere in the grasp.

#	SCENARIO
1	A small, delicate object (an egg, a berry) where excess grip force damages it and only the distal joints should really close.
2	A large, irregular object (a rock, a bundle of cable) that needs the full arm length to wrap around, with the proximal joints doing most of the work.
3	Reaching into a confined gap to retrieve an object, where the arm must stay slender and nearly straight until the very last moment, then curl only at the tip.

There is no single right answer to any of the three. There are answers you can defend and answers you cannot.

6 What to Hand In

ITEM	WHAT IT CONTAINS
Coupling proof	Section 4, full derivation of $\theta_i = f_i(x_1, x_2)$ for every joint.
Kinematic diagram	Labeled schematic (hand-drawn or CAD sketch): all links, joints, tendon routing or differential layout, actuator locations, key dimensions annotated.
Curl sequence plot	Joint angles or overall arm shape at five or more actuator positions from extended to fully wrapped, with a short note on contact order and how you'd tune it further.
CAD assembly	3D assembly in at least two distinct configurations, extended and wrapped, mechanically realizable — no interference, reasonable link thickness and joint clearance.
Physical sketch model	A physical model of the arm, no colour, structure only (cardboard, 3D-printed skeleton, laser-cut links, whatever you have), photographed in at least two configurations, showing the actual coupling mechanism clearly enough to verify against your diagram.
Adapting the design	Section 5, all three object scenarios, with reasoning.
Sources	What you read, and at which point in your process you read it — before or after your own first attempt.

QUESTIONS YOU SHOULD EXPECT

Walk us through your coupling derivation, joint by joint. Why does pulling actuator 1 alone not close every joint equally? What does actuator 2 physically do that actuator 1 cannot? If you added a third joint to the chain without adding a third actuator, what would happen to your coupling relation? Where in the grasp sequence does your arm's actual contact order diverge from what you predicted, and why?

Variable-Damping Suspension for a Sample-Search Rover

A four-wheeled rover searches unmapped, rocky terrain for DNA samples and other biological or chemical evidence. Its chassis carries collection instrumentation that does not tolerate shock loading. The ground is unknown until the moment before a wheel is on it. Design a suspension whose damping changes in real time, driven by terrain sensed just ahead of the rover.

TYPE	DOMAINS	COVERS	TEAM
Cross-domain design	Mechanical lead + Electrical + Software	Semi-active damping, terrain perception, closed loop	3–5 people, see section 0

0 Team Requirement

READ THIS BEFORE YOU START

This problem cannot be solved by one branch working alone, and we will check for that. Your team must be **3 to 5 people**, and across the team you must show real ownership of all three domains: mechanical (the damper and suspension kinematics), electrical (sensing and system architecture), and software (perception and control). A team of five mechanical engineers submitting a plausible-looking control block diagram they cannot explain will be treated the same as a missing section. If your institute does not have enough people to form a natural three-branch team, pair up with another team — a cross-institute submission with real three-domain ownership beats a single-institute submission faking it.

1 Why This Is Hard

Every other problem in this booklet lives inside one domain and can be verified with a calculation or a CAD check. This one cannot be verified that way, because it is not one mechanism — it is a loop.

A camera or LIDAR senses terrain the rover has not reached yet, an embedded system turns that into a terrain estimate, a control algorithm turns the estimate into a target damping coefficient, and a physical damper has to actually achieve that coefficient fast enough to matter before the wheel arrives. If any one link in that loop is wrong — the sensor's field of view does not lead the wheel by enough distance, the controller assumes a damper response time the hardware cannot deliver, the damper's achievable range does not cover the terrain you are driving over — the other two links' good work does not save you. The rover bottoms out anyway.

That is the actual skill being tested. Not can you build a damper, or write a controller, or mount a sensor, but can your team keep those three pieces honest about what the other two need from them.

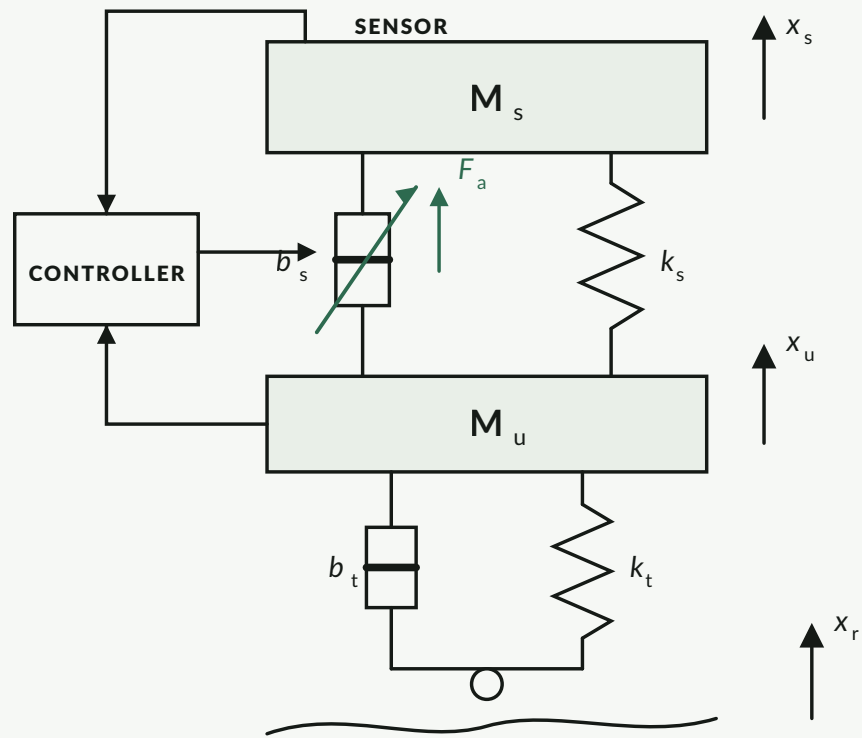


Figure 1. Semi-active quarter-car model. The sprung mass M_s carries the payload; the damper b_s is the variable element, commanded by the controller from sensed terrain.

2 What to Study

Split by domain, because the depth expected differs. Every team member should still be able to explain the other two sections at a basic level — you are one system, not three sub-teams stapled together.

MECHANICAL

- Quarter-car and rocker-bogie suspension models: the mass-spring-damper equations of motion for a wheel and chassis, natural frequency, and damping ratio ζ , and why ζ near critical damping is usually the target for isolating a sensitive payload.
- Semi-active damper technologies: magnetorheological (MR) fluid dampers, electronically controlled variable-orifice dampers, and twin-tube versus mono-tube layouts. What physically changes inside each when you command a different damping coefficient, and how fast it can change.
- Rocker-bogie linkage kinematics: how the geometry redistributes wheel loads over uneven terrain independent of the damper, and where in that linkage a variable damper is actually most effective to place.
- Achievable damping range: how to size your variable damper so its low and high extremes actually bracket the range your control algorithm will ask for. A damper that cannot reach the commanded coefficient is a control-loop failure hiding in the mechanical domain.

ELECTRICAL

- Terrain-sensing hardware: LIDAR versus stereo or depth camera for this use case — range, point density, power draw, and how far ahead of the wheel each can usefully see at rover driving speed.
- Embedded system architecture: partitioning perception, control and actuation across hardware (a single SBC, versus a perception unit talking to a lower-latency microcontroller for the actual damper command), and why the control-loop latency budget drives that decision.
- Actuator interfacing: driving whatever changes your damper's coefficient (a solenoid valve, an MR-fluid current driver, a servo-actuated orifice). Current and voltage requirements, response time, and how you would verify the actuator reaches a commanded state before the wheel gets there.
- Power and packaging constraints typical of a small sample-search rover, and what that rules out before you pick a sensor or actuator.

- Terrain data processing: turning a raw point cloud or depth image into a terrain roughness or obstacle-height estimate ahead of each wheel, at a rate the control loop can use.
- Sensor fusion: combining terrain range data with the rover's own motion (IMU, wheel odometry) to correctly time when a sensed feature will actually reach a given wheel.
- Control strategies for semi-active damping: skyhook and groundhook control, and simpler lookup-table or gain-scheduled approaches mapping terrain roughness to a target damping coefficient. What each assumes and where each breaks down.
- Simulation and verification: modelling your damper and control loop in MATLAB/Simulink before touching hardware, and validating the full sensing-to-actuation pipeline in a rover simulator such as Gazebo.

3 The Objective

Design a variable-damping suspension for a four-wheel rover (rocker or rocker-bogie, your choice, justified) that:

1. Senses terrain ahead of the rover using LIDAR, a depth camera, or an equivalent sensor of your choosing, and produces a terrain roughness or obstacle-height estimate in time to act on it before the corresponding wheel arrives.
2. Feeds that estimate through a defined system architecture — sensor to processing unit to controller to damper actuator — with every interface (data format, rate, latency budget) specified, not just implied by an arrow in a block diagram.
3. Commands a variable damper whose stiffness and/or damping coefficient changes based on that data, targeting a specific damping ratio ζ for the chassis and rocker system that you define and justify.
4. Demonstrates the behaviour, showing the commanded damping coefficient actually changing in response to sensed terrain, and the resulting effect on chassis acceleration or displacement compared to a fixed-damping baseline.

STEP WHAT YOU'RE TESTING

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|----------|--|
| 1 | Model the passive baseline. Derive the quarter-car (or rocker plus chassis) equations of motion with a fixed damper, and compute ζ for a couple of terrain inputs. Show where a fixed damper fails — too harsh here, too soft there. |
| 2 | Choose and size the variable damper. Pick a mechanism and derive its achievable damping-coefficient range against what step 1 showed you actually need. |
| 3 | Design the sensing-to-control pipeline. Specify the sensor, the terrain-estimate algorithm, the control law mapping estimate to target coefficient, and the latency budget end to end — from the point hitting the sensor to the damper physically changing state. |
| 4 | Close the loop in simulation. Run the full pipeline, or a faithful simplified version of it, against a terrain profile with mixed rock sizes. Plot the commanded damping coefficient and the resulting chassis response over time. |
| 5 | Compare against the baseline. Quantify the improvement — peak chassis acceleration, RMS displacement, whatever metric you justify — and be honest about where the variable system does not help or adds risk (latency, actuator wear, added complexity and failure points). |

You are not expected to build physical hardware. A rigorously modelled and simulated system, with every domain's contribution clearly derived and integrated, is the deliverable.

Each is a place error hides if left unstated, and each is a place one domain's assumption can silently break another domain's design.

Name every assumption

Sensor lead distance, loop rate, damper response time, terrain profile tested.

4

The System Architecture Proof

This is the equivalent, for a three-domain system, of the DOF proof in problem 01 and the coupling proof in problem 02. A block diagram alone is not a proof. Show the working.

SHOW THE NUMBERS, END TO END

Starting from your sensor's field of view and update rate, derive the maximum rover speed at which your sensor lead distance still gives the control loop enough time to command the damper *and* the damper enough time to physically reach that state before the sensed feature reaches the wheel. If your rover design exceeds that speed, say so, and say what you would change — sensor placement, control-loop rate, damper response time, or rover speed itself.

5 Adapting the Design

The real test is being asked “what would you change for this ground” and having an answer with reasons behind it, from all three domains at once. For each, say what changes in the damper's target range, the control law, or the sensing setup, and what trade-off that creates elsewhere in the system.

#	SCENARIO
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|---|--|
| 1 | A sudden, isolated large rock strike — a size and shape your terrain sensing might miss or misjudge at speed — that risks shocking the payload directly. |
| 2 | Soft, loose sandy terrain with no hard obstacles, where excess damping stiffness costs traction and ride quality for no protective benefit. |
| 3 | Sustained washboard terrain, many small regular bumps, where the wrong damping setting risks driving the chassis into resonance instead of isolating it. |

There is no single right answer to any of the three. There are answers you can defend, from all three domains together, and answers you cannot.

6 What to Hand In

ITEM	WHAT IT CONTAINS
Team roster & roles	Names and the domains each member owned, matching section 0.
Mechanical package	Rocker-bogie and damper design, equations of motion, ζ derivation for baseline and variable cases, achievable damping range, CAD or scaled sketch of the suspension.
Electrical package	Sensor selection with justification, embedded architecture diagram with every interface labelled with data rate and latency, actuator interfacing and driver design.
Software package	Terrain-processing pipeline, sensor fusion approach, control law (skyhook, ground-hook, gain-scheduled or other, justified), and the simulation code or model used.
Architecture proof	Section 4, full end-to-end latency and speed derivation.
Integrated demonstration	Simulation results showing commanded damping responding to sensed terrain, and chassis response against the fixed-damping baseline. A short video or animation if possible.
Adapting the design	Section 5, all three ground scenarios, with reasoning from all three domains.
Sources	What you read, and at which point in your process you read it.

QUESTIONS YOU SHOULD EXPECT

Walk us through what happens, end to end, in the 200 milliseconds after your sensor detects a rock and before that wheel reaches it. Who on your team can explain the control law, who can explain the damper hardware, and do their explanations agree with each other? What is your damper's actual bandwidth, and where does your control loop assume something faster than that? If your LIDAR or camera missed a rock entirely, what does your system do — and is that answer designed in, or just what happens by default? Where did your fixed-damping baseline actually outperform your variable system, and why?